AI-Native Compliance Infrastructure

Project overview, current status, and what an entrepreneur would build next

October 2026 - prototype stage - built on public data only

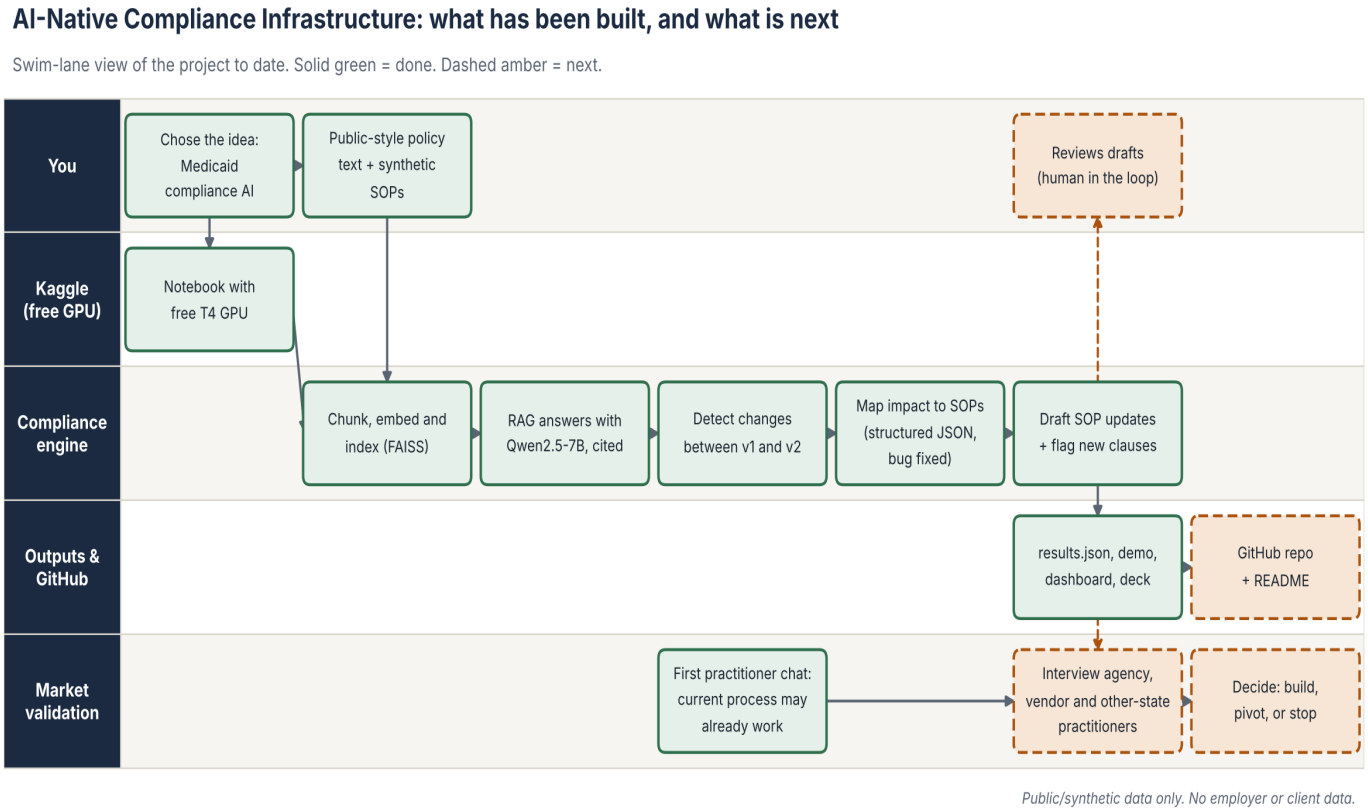
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1. What it is

A tool that reads versions of Medicaid policy documents, detects what changed, works out which internal procedures (SOPs) are affected, and drafts the updated procedure text with citations. Every draft is checked, and any sentence the model invented is flagged for a human reviewer.

Built on a free Kaggle GPU with Qwen2.5-7B, sentence-transformers embeddings and FAISS. No agency, employer or client data is used - only public CMS-style text and synthetic SOPs.

2. What is done



Green = built and tested. Amber dashed = next steps.

Result on the sample change (prior-auth threshold 15 to 20 visits):

- 2 of 3 SOPs correctly flagged as affected; the telehealth SOP correctly left alone (after fixing a string-matching bug).
- The model still adds sentences nobody asked for. A flagger now catches them and shows them to the reviewer instead of silently accepting them.

3. Honest assessment

Question	Answer today
Does the pipeline work end to end?	Yes, on one synthetic change set.
Is it accurate?	Unknown. No labelled test set, so no precision or recall numbers.
Does it work on real documents?	Untested. Real CMS and state manuals are longer and messier.
Will someone pay for it?	Unknown. A first practitioner said the current manual process works and nothing is missed.
Is it defensible?	Not yet. Editing a document with an LLM is a feature, not a company.

The core strategic problem: if the product is only 'rewrite the SOP', the objection 'that is small work' is right. The value has to be what surrounds the rewrite - knowing a change happened, proving it was handled, and keeping an audit trail. That is where the additions below point.

4. What I would add, in priority order

#	Addition	Why it matters	Effort
1	Evaluation harness: 15-30 hand-labelled policy changes, report precision and recall, rerun on every change to the prompts or model	Without numbers you have a demo. With numbers you have evidence for investors and for yourself. This also exposes where the model fails.	1-2 weeks
2	Automatic monitoring of public sources (CMS, Federal Register, State Medicaid Director letters, state bulletins) with alerts	Turns it from a tool you run into a service that runs itself. Works with zero customer data, so you can start today.	2-3 weeks
3	Obligation extraction and deadlines: pull out each 'must', effective date and responsible party; severity rating	'Something changed' is noise. 'You must do X by date Y or risk a penalty' is what a compliance lead acts on.	2 weeks
4	Traceability matrix: policy clause to SOP to owner to status, with audit-ready export	This is the audit evidence. It is the strongest reason to pay, and it is hard to copy because it accumulates customer history.	2-3 weeks
5	Review workflow: approve or reject each draft, tracked-changes Word export, comments, ticket creation (Jira/ServiceNow)	Matches how compliance teams already work. Human sign-off also solves the model-invents-clauses risk.	3 weeks
6	Citation verification: check that every quoted number and clause in a draft appears in the retrieved source text; auto-reject if not	Turns 'we flag invented text' into 'invented text cannot get through'.	1 week
7	Multi-state comparison: same rule across states, side by side	Needs only public data, is sellable as a standalone product to vendors and MCOs operating in several states.	3-4 weeks

#	Addition	Why it matters	Effort
8	Security and deployment story: private cloud deployment, no data retention, audit logs, role-based access; SOC 2 and HIPAA roadmap	Medicaid buyers will not touch a tool without this. Start the design now even if certification comes later.	Ongoing

5. Possible wedges - pick one, do not build all

Wedge	Buyer	Pro	Con
A. Change-intelligence feed (monitoring + summaries + impact tags)	Vendors, MCOs, consultancies	Public data only, fast to launch, easy to test willingness to pay	Competes with existing regulatory-news products
B. SOP impact mapping and drafting (current build)	Compliance teams at plans and vendors	Matches what is built	Needs customer SOPs, so slow trust and sales cycle; the 'small work' objection
C. Contract and RFP requirement tracking: map contract deliverables and modifications to what was delivered	Vendors serving state agencies	Contract changes are expensive and poorly tracked; ties to your PMP and program background	Needs customer documents; narrower market
D. Audit-readiness pack: prove for any rule when it changed, who reviewed, what was updated	Compliance and audit leads	Strongest pain point; recurring need	Needs A and B first

My recommendation: start with A as the front door (zero customer data needed, quick to demo, tests whether anyone cares), use B as the second step for design partners, and grow toward D, which is the defensible product.

6. Business side - what needs answering before more code

- **Customer interviews first.** 5-8 people across agencies, vendors, MCOs and other states. Ask how often a change is missed late, what that cost, and what they use today. Decide build, pivot or stop from what you hear.
- **Competition.** Regulatory-change products already exist in banking and general compliance. Find out who does Medicaid-specific work, what they miss, and why a buyer would switch.
- **Pricing hypothesis.** Test per-state or per-program annual pricing in interviews. Do not guess a number before someone tells you what a missed change costs them.
- **Design partners.** Aim for 2-3 organisations willing to run a free pilot in exchange for feedback and a reference.
- **Cofounder.** You said you want one. The gap to fill is likely sales and domain access on the buyer side, or engineering depth if you stay in the domain-expert role. Decide which you are.
- **Employment and conflict check.** You work in the Medicaid space. Read your employment agreement for outside-work, IP and conflict-of-interest clauses, and get written clearance if needed, before showing this to customers. Keeping the project on public data was the right call; this is the other half.

7. 30 / 60 / 90 day plan

Window	Goal	Done when
Days 1-30	Validate	8 interviews done; 15 labelled changes; accuracy numbers; decision on wedge
Days 31-60	Make it real	Runs on real public versioned documents; monitoring and obligation extraction working; citation verification in place
Days 61-90	Prove it	One design partner pilot or 3 written letters of interest; traceability matrix demo; updated deck

8. Risks

- **No real demand.** Mitigation: interviews before engineering.
- **Model errors in a high-stakes domain.** Mitigation: human review, citation checks, accuracy reporting.
- **Trust and data access.** Mitigation: private deployment, public-data wedge first.
- **Incumbents.** Mitigation: Medicaid specialisation, audit trail as the moat.
- **Founder time.** Mitigation: cofounder, tight scope, one wedge.

Repo: github.com/shubhamkumarsingh91-lgtm/Compliance-AI. No proprietary, client or employer data is used in this project.